LEVEL 18 · PART VIII

Platform Engineering 18

Goal: Build internal platforms that let product teams self-serve infrastructure safely and quickly.

RECOMMENDED TIME
4-6 weeks

TOPICS
10

PRACTICE / CAPSTONE
Build a mini internal developer platform: Backstage catalog + software template that scaffolds a service repo, provisions infra via Crossplane/Terraform and deploys through Argo CD.

Topics

#	Topic	What you will learn
1	Internal Developer Platforms **p.3405**	IDP concept, platform as a product, thinnest viable platform
2	Developer Portals **p.3415**	what a portal offers, build vs buy, adoption
3	Self-service Infrastructure **p.3427**	request flows, guardrails, ticket-ops to API-ops
4	Golden Paths **p.3439**	paved roads vs mandates, templates, measuring adoption
5	Service Catalogs **p.3451**	ownership metadata, catalog-info.yaml, dependency graphs, scorecards
6	Infrastructure Templates **p.3463**	reusable modules/compositions, versioning, template testing
7	Policy as Code **p.3475**	OPA/Rego, Kyverno, Conftest, enforcing in CI and cluster
8	Backstage **p.3491**	install, catalog, software templates, plugins, TechDocs
9	Crossplane **p.3505**	providers, managed resources, compositions, XRDs, claims
10	Integrating Argo CD, Terraform and Kubernetes into a Platform **p.3521**	how the pieces compose: control planes, GitOps delivery, state ownership

#	Topic	What you will learn
★	Level capstone project **p.3535**	Build a mini internal developer platform: Backstage catalog + software template that scaffolds a service repo, provisions infra via Crossplane/Terraform and deploys through Argo CD.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

